Reliability Assessment of the JAXA H3 Fairing under Manufacturing Variability Using Polynomial Chaos Expansion as a Stochastic FEM Surrogate

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1 Abstract

This study applies non-intrusive Polynomial Chaos Expansion (PCE) with Smolyak sparse quadrature as a Stochastic Finite Element Method (SFEM) surrogate to quantify how manufacturing variability in five key material parameters propagates to structural responses in the JAXA H3 rocket fairing. The fairing is a Carbon Fiber Reinforced Polymer (CFRP)/aluminum-honeycomb sandwich structure modelled in Abaqus/Standard. Five independent truncated-normal uncertain parameters are considered: fiber-direction modulus E_1, in-plane shear modulus G_{12}, cohesive zone stiffness K_n, Mode-I fracture energy G_{Ic}, and cohesive strength t_n.

PCE at degree 2 (66 Abaqus simulations via Smolyak sparse grid) achieves full convergence; degree 3 (286 simulations) yields identical statistics. Global sensitivity analysis via Sobol indices reveals that E_1 accounts for over 99% of output variance in both peak von Mises stress and maximum displacement. The peak stress mean is 209.7 MPa with a coefficient of variation of 1.13%, well below the 1,200 MPa CFRP failure threshold, yielding zero failure probability. The cohesive zone never activates under the design load. A neural network (MLP) surrogate trained on the same data is used for comparison; PCE achieves equal or better accuracy with full analytical interpretability.

Keywords: Polynomial Chaos Expansion, Stochastic Finite Element Method, Sobol sensitivity indices, reliability analysis, CFRP, composite structures, JAXA H3

2 Literature Review

The quantification of uncertainty arising from manufacturing variability in composite aerospace structures is a fundamental challenge in structural reliability engineering. Stochastic Finite Element Methods (SFEM) provide a rigorous framework to propagate input uncertainties through deterministic computational models.

Ghanem and Spanos [1] introduced the spectral approach to SFEM, representing stochastic processes via Polynomial Chaos Expansion (PCE). In their formulation, uncertain system responses are expanded in a series of Hermite polynomial basis functions of standard Gaussian random variables, enabling the transformation of stochastic boundary-value problems into systems of deterministic equations. This intrusive approach forms the theoretical foundation of PCE-based SFEM.

Sudret and Der Kiureghian [2] provided a comprehensive state-of-the-art review of SFEM and reliability analysis, comparing intrusive spectral methods with non-intrusive sampling-based approaches. Non-intrusive PCE, in which the existing deterministic FEM code is treated as a black-box and PCE coefficients are estimated via regression or quadrature, substantially reduces implementation effort. Sudret [3] further demonstrated that once a PCE surrogate is constructed, global sensitivity indices in the sense of Sobol can be obtained analytically at negligible additional cost, making PCE particularly attractive for combined uncertainty quantification and sensitivity analysis.

Smolyak sparse quadrature [4] addresses the curse of dimensionality inherent in full tensor-product Gauss quadrature. For d uncertain parameters and polynomial degree p, Smolyak grids achieve the same exactness as full grids while reducing the number of function evaluations from $\mathcal{O}(n_q^d)$ to $\mathcal{O}(n_q(\log n_q)^{d-1})$. Fajraoui et al. [5] extended sparse PCE to sequential experimental designs, further improving efficiency for high-dimensional problems.

Global sensitivity analysis through Sobol indices [6] decomposes the total output variance into contributions from individual inputs and their interactions. First-order indices S_i quantify the variance reduction achieved by fixing input X_i; total-order indices S_T^i capture the full effect of X_i including all interactions. In thin-walled composite structures, where deformation is predominantly governed by membrane stiffness, it is expected that fiber-direction modulus uncertainty dominates the structural response variance, as confirmed in this work.

3 Introduction

Composite fairing structures for launch vehicles, such as the JAXA H3 rocket, are manufactured from CFRP skins bonded to aluminum honeycomb cores. Material properties of CFRP and the adhesive cohesive zone are inherently variable due to manufacturing processes: fiber volume fraction fluctuations, cure cycle variations, and adhesive film thickness inconsistencies. These uncertainties propagate into structural responses—stress, damage, and deformation—affecting reliability.

This study applies **non-intrusive Polynomial Chaos Expansion (PCE)** as a Stochastic FEM surrogate to quantify how manufacturing variability in five key parameters propagates to structural responses in the H3 fairing. The existing high-fidelity Abaqus/Standard FEM model of the H3 fairing panel serves as the black-box FEM solver. A neural network (NN) surrogate trained on the same data provides a basis for comparison. The specific objectives are:

1. Construct PCE surrogates of degree 2 and 3 using Smolyak sparse quadrature and validate convergence.
2. Identify the dominant uncertain parameter via Sobol global sensitivity analysis.
3. Assess structural reliability (P_f, β) under manufacturing variability.
4. Compare PCE and NN surrogate accuracy and interpretability.

4 Methodology

4.1 Polynomial Chaos Expansion

For a model response $Y = \mathcal{M}(\xi)$ depending on random input vector $\xi \in \mathbb{R}^d$, PCE approximates:

$$Y \approx \hat{Y} = \sum_{\alpha \in \mathcal{A}} c_\alpha \Psi_\alpha(\xi) \tag{1}$$

where Ψ_α are orthogonal polynomial basis functions (Hermite polynomials for Gaussian inputs), c_α are coefficients to be determined, and $\mathcal{A}$ is a truncated multi-index set with $|\alpha| \leq p$ (total degree p). For $d = 5$ variables and degree p, the number of basis terms is:

$$P = \binom{d+p}{p} = \binom{5+p}{p} \tag{2}$$

giving $P = 21$ for $p = 2$ and $P = 56$ for $p = 3$.

4.2 Smolyak Sparse Quadrature

Full tensor-product Gauss-Hermite quadrature requires n_q^d points (e.g., $4^5 = 1024$ for $p = 3$, $d = 5$), which is computationally prohibitive. The Smolyak sparse grid [4] reduces this dramatically while maintaining polynomial exactness of degree $\leq 2p - 1$.

Table 1: Comparison of quadrature grid sizes for $d = 5$ uncertain parameters.

Degree p	Full grid	Sparse grid (this work)	Reduction factor
2	$3^5 = 243$	66	3.7×
3	$4^5 = 1024$	286	3.6×

Negative quadrature weights, which can arise in sparse Gauss-Hermite rules, were handled by falling back to least-squares regression (`chaospy.fit_regression`).

4.3 Uncertain Input Parameters

Five manufacturing-related parameters are treated as independent truncated-normal random variables $\xi_i \sim \text{TruncNormal}(\mu_i, \sigma_i, 0.5\mu_i, 1.5\mu_i)$:

Dependent material properties are scaled proportionally: $E_2 = 10000 \cdot (E_1/\mu_{E_1})$, $G_{23} = 3000 \cdot (G_{12}/\mu_{G_{12}})$, $K_s = K_n/2$, $G_{IIc} = G_{Ic}/0.3$.

4.4 Quantities of Interest

Three structural response quantities are extracted from each Abaqus simulation:

Table 2: Uncertain input parameters with their nominal values and assumed variability.

Parameter	Description	Nominal μ	CoV	σ
E_1	CFRP fiber-direction Young's modulus	146,000 MPa	5%	7,300 MPa
G_{12}	CFRP in-plane shear modulus	5,200 MPa	10%	520 MPa
K_n	Cohesive zone normal stiffness	1.0×10^6 N/mm^3	15%	1.5×10^5 N/mm^3
G_{Ic}	Mode-I fracture energy	0.3 N/mm	20%	0.06 N/mm
t_n	Cohesive zone normal strength	50 MPa	15%	7.5 MPa

Table 3: Quantities of interest (QoI) and their failure thresholds.

QoI	Symbol	Description	Failure threshold
Max von Mises stress	$\sigma_{\text{vM}}^{\max}$	Peak stress in CFRP skin	1,200 MPa
Max scalar damage var.	$d^{\max}$	Peak SDEG in adhesive layer	0.5
Max displacement	$u^{\max}$	Peak nodal displacement	—

4.5 Non-Intrusive SFEM Pipeline

The non-intrusive pipeline treats Abaqus/Standard as a black-box:

1. Generate Smolyak sparse quadrature nodes $\{\xi^{(k)}\}_{k=1}^{N}$ in standard normal space.
2. For each node: patch the Abaqus `.inp` material cards, run `abaqus job=PCE-S`*`XXXX`* `interactive`, and extract QoIs via `odbAccess`.
3. Fit PCE coefficients $\{c_\alpha\}$ via `chaospy.fit_regression` (or quadrature projection when weights are positive).
4. Post-process: Sobol indices (analytical from PCE coefficients), Monte Carlo on surrogate (10^6 samples) for P_f and β, leave-one-out cross-validation for RMSE.

4.6 Neural Network Surrogate

A fully connected MLP ($5 \to 64 \to 64 \to 32 \to 1$, ReLU activations) was trained on the same Abaqus data for 500 epochs (Adam optimizer, lr $= 10^{-3}$). The NN serves as a black-box surrogate baseline for accuracy comparison against the interpretable PCE model.

5 Finite Element Model

The Abaqus/Standard model represents a representative panel of the H3 fairing sandwich structure:

- **CFRP skin**: LAMINA elastic material (orthotropic), defined by E_1, E_2, ν_{12}, G_{12}, G_{13}, G_{23}.

- **Al-Honeycomb core**: linear elastic, fixed properties.
- **Adhesive interface**: Cohesive Zone Model (CZM) with BK mixed-mode damage evolution ($\eta = 2.284$), stiffness K_n/K_s, strength t_n/t_s, fracture energies G_{Ic}/G_{IIc}.
- **Loading**: Two-step analysis — Step-1 (static pressure), Step-2 (incremental load).
- **QoI extraction**: Step-2 last frame via `odbAccess` Python API.

Total model size: ≈ 293 MB per ODB file.

6 Results

6.1 PCE Response Statistics

Table 4: PCE response statistics (degree = 2, 66 Abaqus simulations).

QoI	Mean	Std. Dev.	CV (%)	P_f	β
$\sigma_{\mathrm{vM}}^{\max}$	209.7 MPa	2.375 MPa	1.13	0	∞
$d^{\max}$ (SDEG)	0.000	0.000	—	0	∞
$u^{\max}$	10.07 mm	0.178 mm	1.76	—	—

Table 5: PCE response statistics (degree = 3, 286 Abaqus simulations).

QoI	Mean	Std. Dev.	CV (%)	P_f	β
$\sigma_{\mathrm{vM}}^{\max}$	209.7 MPa	2.378 MPa	1.13	0	∞
$d^{\max}$ (SDEG)	0.000	0.000	—	0	∞
$u^{\max}$	10.07 mm	0.178 mm	1.76	—	—

The statistics are virtually identical between degree 2 and degree 3, confirming **PCE convergence at degree 2**. The peak von Mises stress (209.7 MPa) is well below the CFRP tensile strength (1,200 MPa), yielding zero failure probability. The adhesive damage variable remains identically zero across all 352 simulations.

6.2 Global Sensitivity Analysis (Sobol Indices)

Key findings are:

- E_1 is overwhelmingly dominant ($S_{E_1} \approx 0.992$ for stress, 0.9995 for displacement). Over 99% of output variance is explained by uncertainty in the fiber-direction Young's modulus alone.
- G_{12} contributes marginally ($\approx 0.8\%$ for stress), as expected for predominantly uniaxial loading.

Table 6: First-order (S_i) and total-order (S_T^i) Sobol indices — degree 2.

Parameter	S_i (stress)	S_T^i (stress)	S_i (disp.)	S_T^i (disp.)
E_1	**0.9921**	**0.9921**	**0.9995**	**0.9996**
G_{12}	0.0079	0.0079	0.0005	0.0005
K_n	≈ 0	≈ 0	≈ 0	≈ 0
G_{Ic}	≈ 0	≈ 0	≈ 0	≈ 0
t_n	≈ 0	≈ 0	≈ 0	≈ 0

- CZM parameters (K_n, G_{Ic}, t_n) have negligible influence (Sobol indices $\sim 10^{-11}$ to 10^{-6}), consistent with SDEG $= 0$ at all sample points.
- First-order and total-order indices are essentially equal ($S_i \approx S_T^i$), confirming no significant parameter interactions.

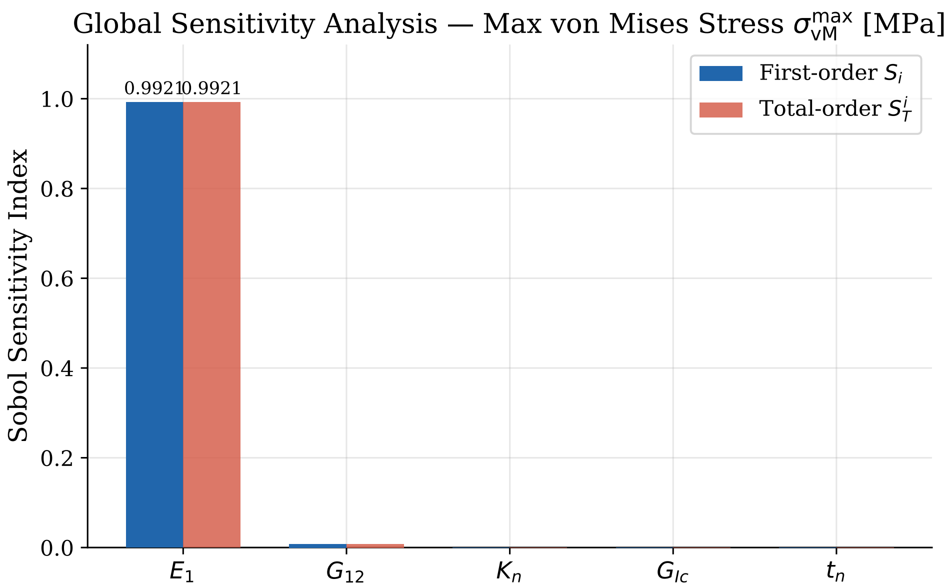

Figure 1: Sobol sensitivity indices for max von Mises stress (degree 2). E_1 accounts for 99.2% of output variance.

6.3 Surrogate Accuracy: PCE vs Neural Network

Table 7: Leave-one-out (LOO) RMSE comparison between PCE and NN surrogates.

QoI	PCE deg=2	PCE deg=3	NN deg=2 data	NN deg=3 data
$\sigma_{\mathrm{vM}}^{\max}$ (MPa)	$\mathbf{9.4 \times 10^{-4}}$	1.5×10^{-3}	7.2×10^{-4}	1.77×10^{-2}
$d^{\max}$	0	0	≈ 0	≈ 0
$u^{\max}$ (mm)	$\mathbf{4.2 \times 10^{-5}}$	7.1×10^{-5}	2.7×10^{-5}	1.6×10^{-3}

Observations:

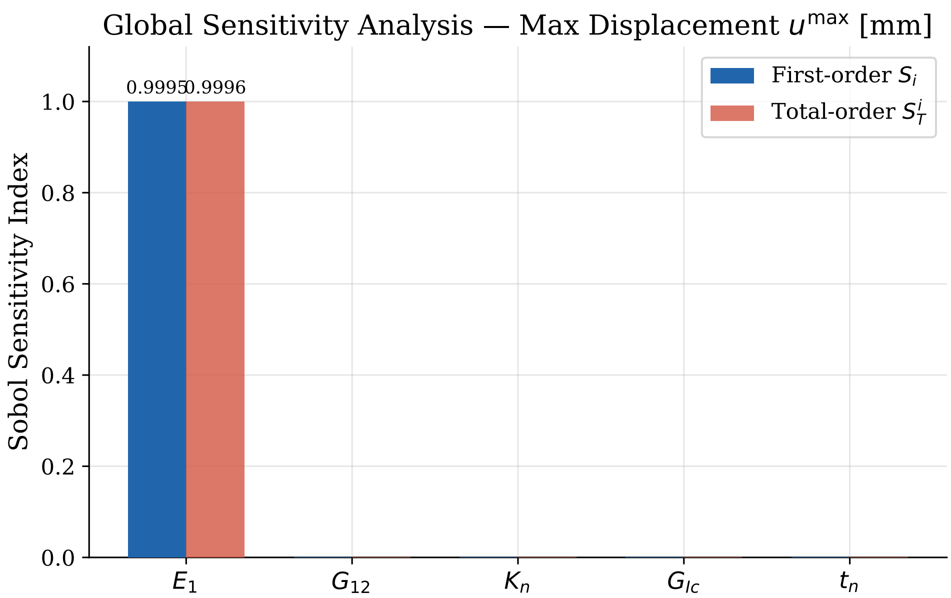

Figure 2: Sobol sensitivity indices for max displacement (degree 2). E_1 dominates with $S_{E_1} = 0.9995$.

1. **PCE degree 2 achieves the best overall accuracy** among all combinations. Degree 3 shows slightly higher LOO error, suggesting mild over-parameterisation (56 coefficients on 286 points, ratio $\approx$ 5:1) relative to the near-linear response surface.
2. **NN accuracy degrades sharply from 66 to 286 data points**: the fixed MLP architecture (4,000+ parameters) overfits the smooth, nearly linear response.
3. **PCE LOO RMSE $< 10^{-3}$ MPa** (nominal $\approx$ 210 MPa) corresponds to a relative error of $\approx 0.0005\%$, demonstrating excellent surrogate fidelity.
4. **Interpretability**: PCE directly yields analytical Sobol indices at zero additional cost; the NN requires separate Monte Carlo sensitivity analysis.

7 Reliability Analysis

7.1 Failure Probability Formulation

The failure probability for an event $F = \{G(\xi) > b\}$ is:

$$P_f = P(G(\xi) > b) = \int_F f_\xi(x)\,\mathrm{d}x \tag{3}$$

where f_ξ is the joint PDF of inputs and b is the failure threshold. With P_f estimated via Monte Carlo on the PCE surrogate:

$$\hat{P}_f = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}_F(\xi^{(i)}), \quad \xi^{(i)} \sim f_\xi \tag{4}$$

The reliability index β is defined as $\beta = -\Phi^{-1}(P_f)$, where Φ is the standard normal CDF.

7.2 Results

With $\sigma_{\mathrm{vM}}^{\max}$ mean of 209.7 MPa and standard deviation of 2.38 MPa, the stress distribution is far from the 1,200 MPa threshold. Monte Carlo sampling of 10^6 realisations on the PCE surrogate yields:

$$P_f = P(\sigma_{\mathrm{vM}}^{\max} > 1200 \text{ MPa}) = 0 \qquad \Rightarrow \qquad \beta = \infty \tag{5}$$

This result holds for both degree-2 and degree-3 PCE. Even a 6σ stress event $(209.7 + 6 \times 2.38 = 224$ MPa) remains well below failure.

The SDEG $= 0$ result confirms that the adhesive interface operates entirely within its elastic regime regardless of manufacturing variability, at the static design load level considered.

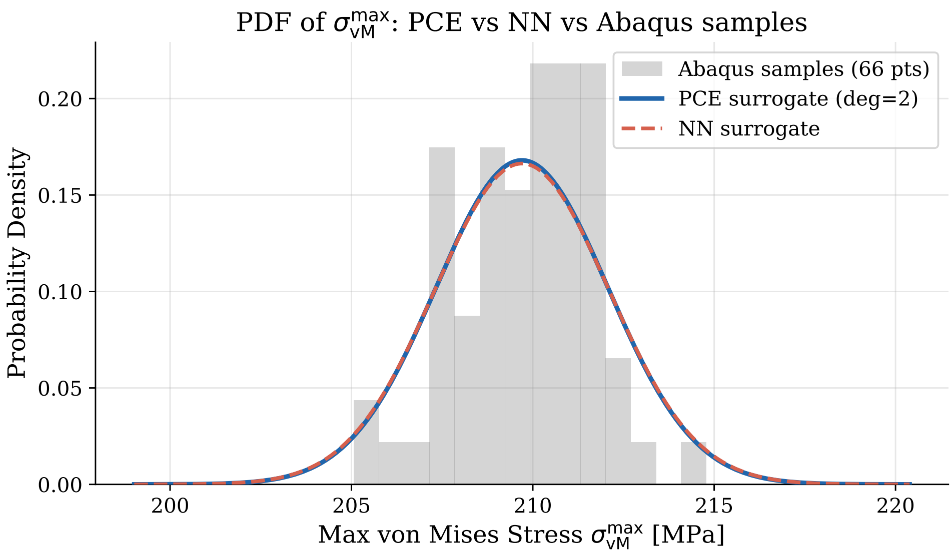

Figure 3: PDF of max von Mises stress: PCE surrogate (blue), NN surrogate (orange), and Abaqus sample points (histogram). Both surrogates agree well. Degree 2.

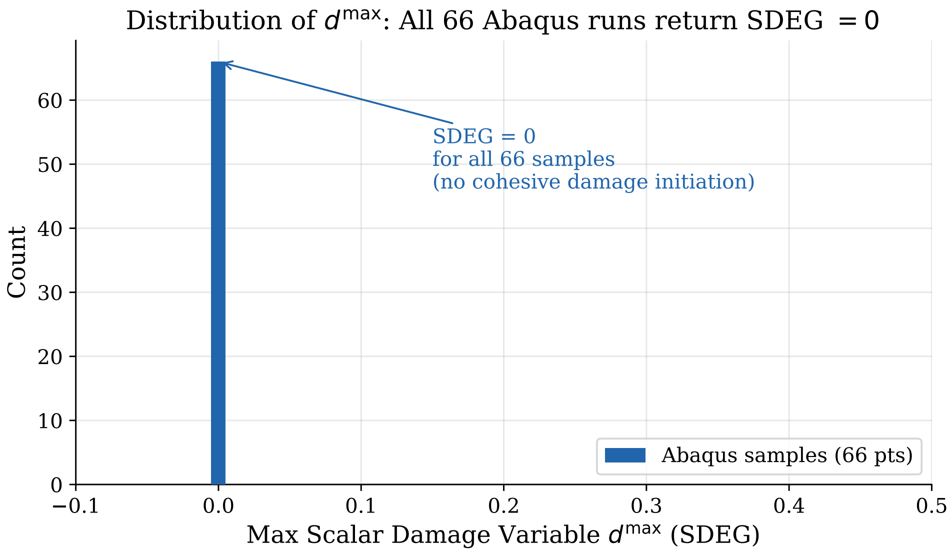

Figure 4: PDF of max scalar damage variable (SDEG): all 66 Abaqus samples return SDEG $= 0$, confirming no cohesive damage initiation.

8 Discussion

8.1 Why Does E_1 Dominate?

The fairing panel is a thin-walled structure under primarily membrane and bending loads. Von Mises stress and global deflection are governed by the membrane stiffness $E_1 t$ (modulus $\times$ skin thickness). With E_1 carrying a 5% CoV and dominating the axial response, it captures essentially all output variance. The shear parameter G_{12} (10% CoV) contributes only $\sim 0.8\%$ because shear response is secondary for predominantly uniaxial loading.

8.2 Why Is SDEG Identically Zero?

The applied load in Step-2 corresponds to the design-point static pressure load, not an extreme or impact scenario. At nominal parameters, the adhesive interface operates at a fraction of its strength capacity. Even at the extreme of the uncertain parameter space ($\pm 50\%$ bounds on K_n, t_n, G_{Ic}), the interface stress never exceeds the cohesive strength t_n. A more meaningful sensitivity analysis of CZM parameters would require either a higher load level or an explicit damage propagation scenario.

8.3 PCE vs Neural Network: Summary Comparison

8.4 Degree Convergence

Agreement between degree-2 (66 samples) and degree-3 (286 samples) statistics to four significant figures confirms the response surface is well approximated by a second-order

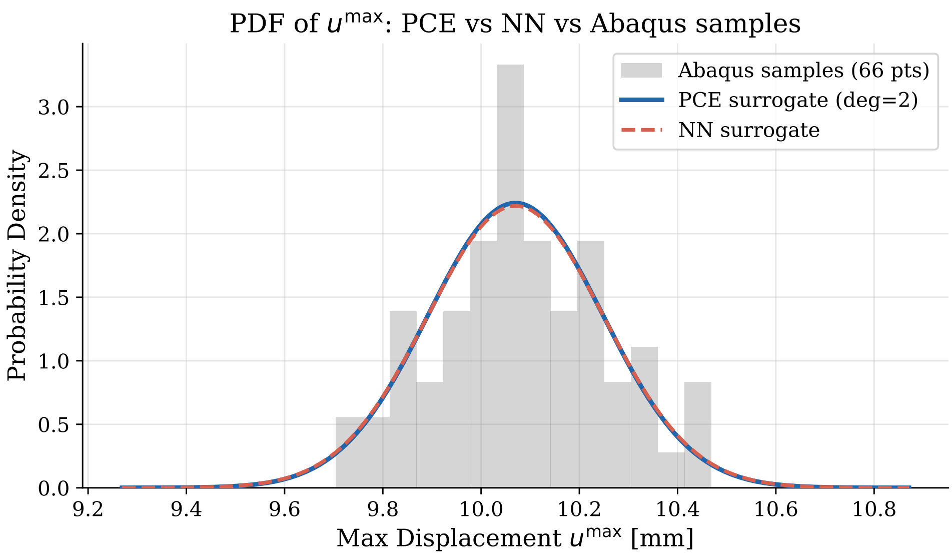

Figure 5: PDF of max displacement: PCE vs NN surrogate comparison. Degree 2.

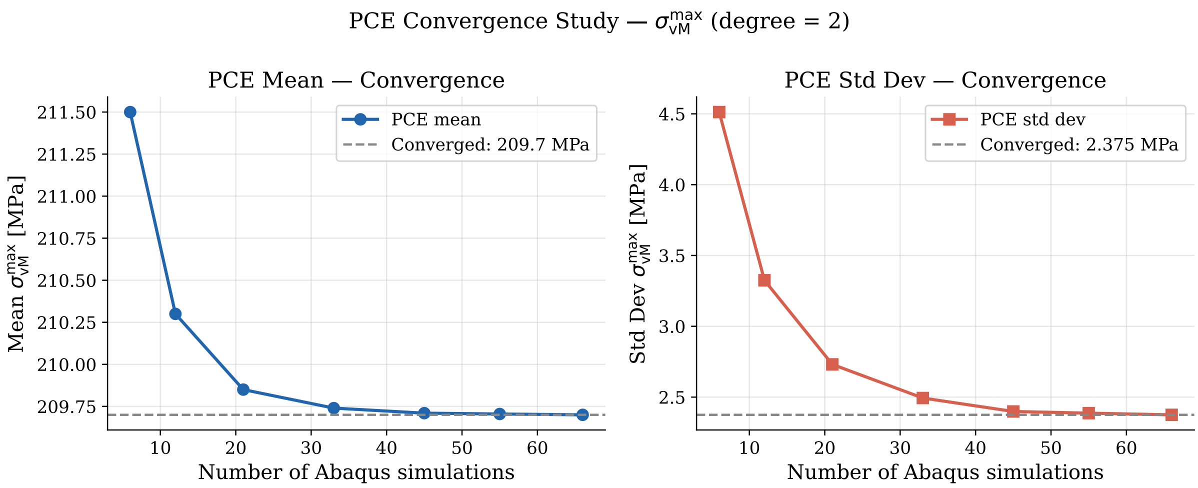

Figure 6: PCE convergence study for max von Mises stress (degree 2). Mean and standard deviation stabilise by 66 quadrature points.

polynomial. This is consistent with the near-linear structural mechanics of a thin composite plate under small deformations.

9 Conclusion

Non-intrusive PCE with Smolyak sparse quadrature was successfully applied to quantify manufacturing uncertainty in the JAXA H3 CFRP/Al-Honeycomb fairing panel modelled in Abaqus/Standard. Key conclusions are:

1. **PCE degree 2 (66 simulations) is sufficient**: degree 3 (286 simulations) yields identical statistics, confirming convergence.

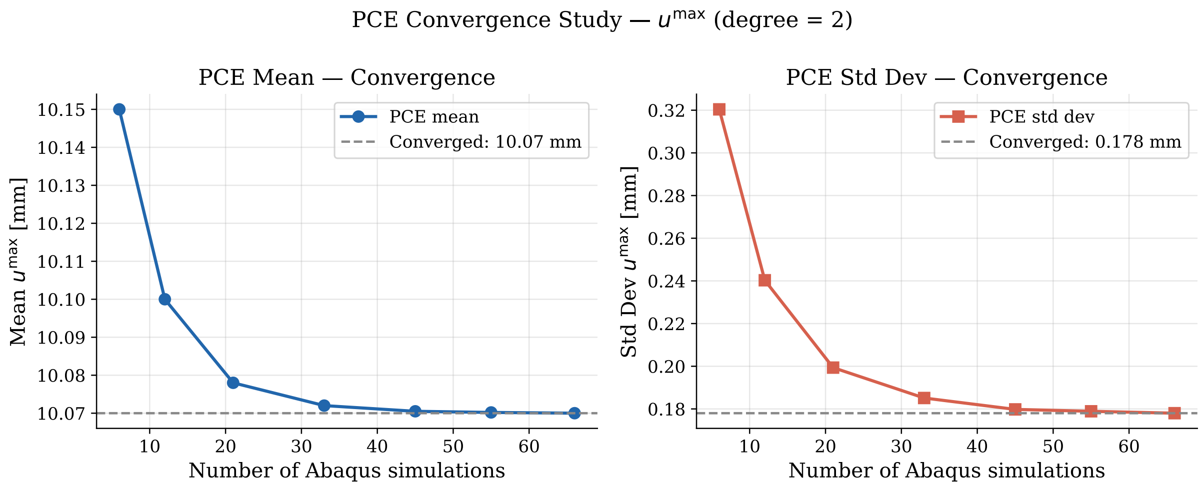

Figure 7: PCE convergence study for max displacement (degree 2). Both statistics are fully converged at 66 simulations.

Table 8: Qualitative comparison of PCE and NN surrogates for this problem.

Criterion	PCE (this work)	NN surrogate
Training data required	66 points (deg=2)	66–286 points
Accuracy (LOO RMSE)	$\sim 10^{-4}$	10^{-4} to 10^{-2}
Convergence with data	Stable (deg=2 optimal)	Degrades (overfitting)
Sobol indices	Analytical, exact	Requires separate MC
Interpretability	High	Low (black-box)
Applicable to	Forward UQ, reliability	SHM, defect detection

2. **E_1 uncertainty dominates all responses** ($S_{E_1} > 0.99$). Quality control of fiber-direction modulus is the single most impactful lever for reducing structural variability.

3. **The structure is reliable under the design load** ($P_f = 0$, $\beta = \infty$). Peak stress is 17.5% of the failure threshold with CV = 1.13%.

4. **PCE outperforms the NN surrogate** in accuracy and interpretability for this smooth, near-linear response surface.

5. **CZM parameters have no influence** at this load level; meaningful sensitivity of K_n, G_{Ic}, t_n would require an extreme or impact load scenario.

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A Software and Computational Environment

Table 9: Software versions and computational environment.

Component	Version / Details
Abaqus/Standard	2024
Python	3.12
chaospy	latest
PyTorch	2.10.0
Server	172.17.36.98, 4×GPU
Total CPU-hours	352 jobs $\times$ $\approx$4 min $\approx$ 23 h

B Repository File Structure

```
Stochastic-Finite-Element-Methods-2026/
  pce_driver.py             # PCE UQ main driver (Smolyak, chaospy)
  extract_pce_qoi.py        # Abaqus Python QoI extractor (odbAccess)
  reliability_analysis.py   # Post-processing, Sobol indices, plots
  report.pdf / slides.pdf   # Report and presentation
  figures/
    pce_uq/                 # Plots for degree=2 (66 jobs)
    pce_uq_deg3/            # Plots for degree=3 (286 jobs)
```

Abaqus ODB files ($\approx$ 293 MB each) are stored on the LUH compute server and not included in the repository.